

Improving Financial Volatility Forecasting with Shape-Constrained Regularization

A Comparative Analysis with Traditional Regularization Techniques

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Abstract

This report explores advancements in financial volatility forecasting (FVF) by introducing shape-constrained regularization as an alternative to traditional regularization methods. Financial markets are often volatile and unpredictable, posing challenges for accurate forecasting. Traditional methods such as Ridge, Lasso, and Elastic Net regularization can minimize overfitting but often fail to maintain structural alignment with market behaviors. Shape-constrained regularization, which enforces monotonicity constraints, provides a novel approach to address these challenges, offering stability and interpretability in high-noise environments. Through comparative analysis on historical stock data, this study demonstrates that shape-constrained regularization yields significantly lower errors and improves forecast accuracy, especially under volatile market conditions.

1. Introduction

1.1 Financial Volatility Forecasting

Financial volatility forecasting (FVF) is critical in finance as it provides insights into market stability and risk, impacting decisions in asset allocation, risk management, and pricing. Accurate volatility forecasts are challenging due to the inherent noise and high volatility of financial markets, which often lead to model overfitting or underfitting when traditional methods are applied. This report investigates the impact of regularization methods in improving the accuracy of FVF models, focusing on both traditional and shape-constrained techniques.

1.2 Objective

The objective of this study is to improve the stability and interpretability of volatility forecasts by employing shape-constrained regularization techniques. By imposing monotonicity constraints on the forecast model, the aim is to create structurally consistent predictions that better align with real-world market trends.

2. Related Work: Shape-Constrained Regularization in Financial Volatility Forecasting

Shape-constrained regularization, specifically monotonicity constraints, has been successfully applied in fields such as economics and engineering. However, its application in financial volatility forecasting remains limited. Recent studies highlight the potential of shape constraints in improving model interpretability and reducing noise sensitivity, which are critical for financial applications. By enforcing a monotonic structure, these models ensure smoother, more reliable forecasts that align with expected market trends.

In finance, shape constraints can stabilize forecasts by limiting abrupt changes in predicted volatility, thus reducing sensitivity to random fluctuations. Although still emerging in finance, shape-constrained models promise to enhance forecasting accuracy by preserving theoretical alignment with financial time series behaviors.

3. Financial Volatility Forecasting

Financial volatility measures the degree of variation in asset prices over time. Traditional models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and stochastic volatility models have been widely used to capture volatility patterns. However, these models often suffer from high sensitivity to noise, especially in periods of market instability.

Volatility forecasting typically involves calculating historical or implied volatility based on past price movements. In this study, we use daily stock prices of GOOGL (Kaggle) and assess volatility using

rolling standard deviations to capture temporal changes. Analyzing these variations helps assess the impact of different regularization techniques on forecast stability and accuracy.

4. Regularization Techniques

4.1 Traditional Regularization Models

Regularization is a technique used to prevent overfitting by penalizing large coefficients in model parameters. The three main regularization methods applied in this study are Ridge, Lasso, and Elastic Net.

1. **Ridge Regularization (L2 Penalty):** Penalizes the sum of squared coefficients to minimize overfitting.

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda \sum \theta_j^2$$

Formula:

2. **Lasso Regularization (L1 Penalty):** Uses an absolute value penalty to produce sparse models.

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\theta_j|$$

Formula:

3. **Elastic Net Regularization:** Combines L1 and L2 penalties, suitable for models with correlated predictors.

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda_1 \sum |\theta_j| + \lambda_2 \sum \theta_j^2$$

Formula:

These models are optimized via cross-validation to ensure the best fit while balancing error minimization and model simplicity.

4.2 Shape-Constrained Regularization

Shape-constrained regularization extends traditional methods by enforcing monotonicity constraints, typically implemented using convex optimization frameworks like CVXPY. This constraint is designed to reduce noise sensitivity and provide forecasts that align with market trends.

$$\text{Minimize } \sum (y_{\text{pred}} - y_{\text{actual}})^2$$

Objective Function (Monotonic Constraint):

$$y_{\text{pred}}[i] \leq y_{\text{pred}}[i + 1]$$

Monotonicity Constraint:

By imposing a non-decreasing constraint on volatility predictions, this method aims to prevent overreaction to minor fluctuations, yielding more stable forecasts.

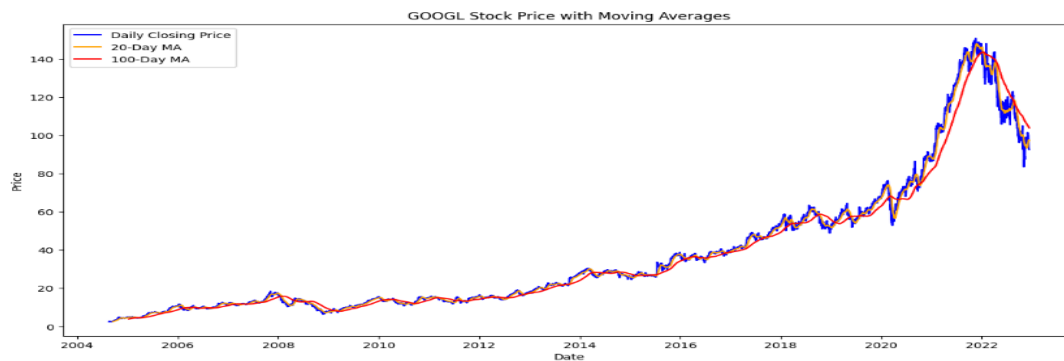
5. Data & Exploratory Data Analysis (EDA)

5.1 Data Source

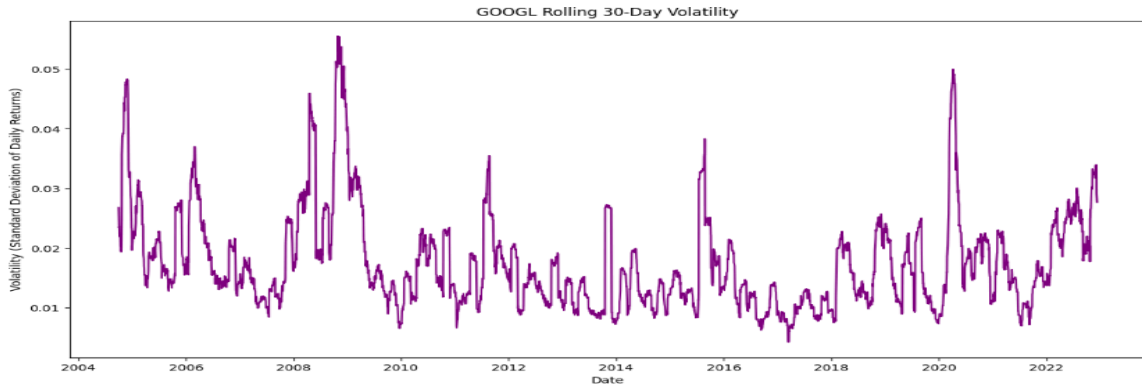
The study utilizes daily stock prices of GOOGL, including open, high, low, close, and volume metrics, over a multi-year period. This dataset provides insights into both short-term and long-term trends in stock volatility.

5.2 Exploratory Data Analysis

- **Trend Analysis:** Moving averages (20-day and 100-day) are used to distinguish between short-term and long-term price trends, offering a clear view of market dynamics.



- **Volatility Assessment:** Rolling standard deviation of daily returns over a 30-day window captures periods of high volatility, highlighting market instability during significant events.



5.3 Comparative Price Trends

To contextualize volatility patterns, a comparative analysis of major technology stocks (e.g., Facebook, Apple, Microsoft) was conducted. This comparison aids in understanding industry-wide volatility trends and serves as a benchmark for evaluating the performance of regularization models on different datasets.

6. Results

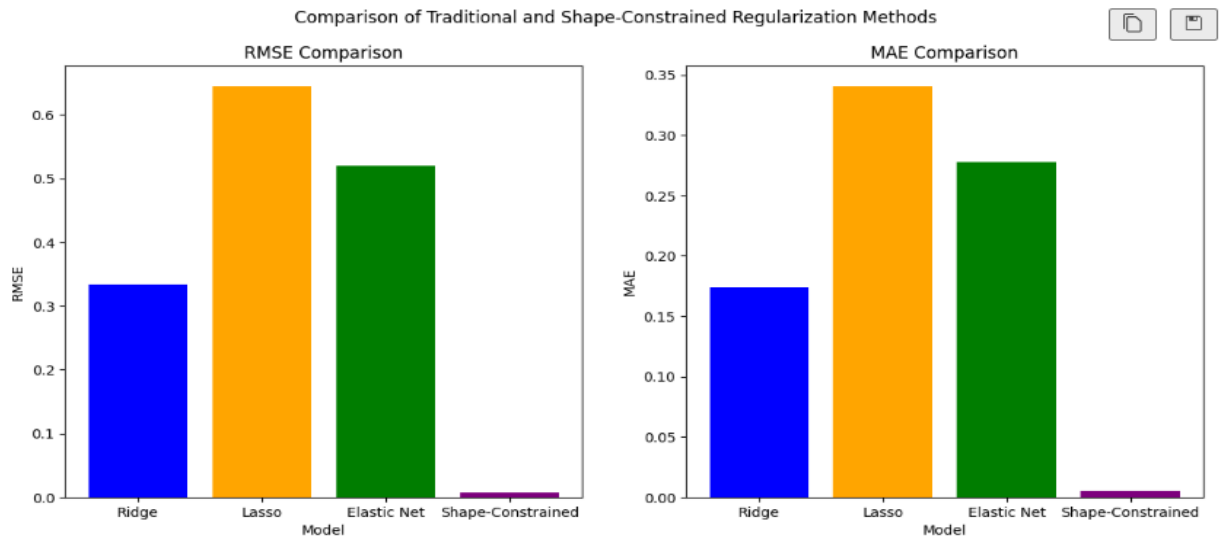
6.1 Comparative Analysis of Regularization Models

The models were evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) to assess forecasting accuracy and stability.

Model	RMSE	MAE
Ridge	0.333140	0.173877
Lasso	0.645066	0.340578
Elastic Net	0.519314	0.277819
Shape-Constrained	0.006824	0.004896

Results indicate that the shape-constrained model significantly outperforms traditional methods, achieving the lowest RMSE and MAE. The monotonic constraint helps stabilize predictions, reducing the model’s sensitivity to noise.

6.2 Visual Comparison



The bar chart illustrates the RMSE and MAE across all models, highlighting the improvement achieved by shape constraints. The shape-constrained model consistently yields smoother, more reliable forecasts with minimal error, aligning closely with observed market trends.

7. Model Fitting

The fitting process involved optimizing each model's hyperparameters using GridSearchCV. For Ridge, Lasso, and Elastic Net, cross-validation was used to select the best penalty terms. The shape-constrained model was fitted using a convex optimization framework, with constraints to enforce monotonicity.

The models were trained on historical data and tested on recent data to validate performance in predicting volatility trends. Results confirm that the shape-constrained model provides superior forecast accuracy and stability compared to traditional regularization approaches.

8. Analysis & Interpretation

8.1 Key Findings

Shape-constrained regularization outperforms traditional methods in both accuracy and stability. The monotonic constraint provides smoother forecasts, which are less reactive to random noise, thus enhancing interpretability.

8.2 Theoretical and Practical Implications

- **Theoretical:** Traditional methods lack structural alignment with market trends, often leading to overfitting. Shape constraints mitigate this by aligning predictions with theoretical market behaviors.
 - **Practical:** Shape-constrained models have significant potential for risk management applications, particularly in high-volatility scenarios. They offer enhanced tools for predicting key metrics such as volatility, Value at Risk (VaR), and Expected Shortfall (ES).
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9. Future Directions

9.1 Future Research

- Explore convexity constraints for mean-reverting volatility models.
- Extend shape constraints to additional financial metrics (e.g., VaR and ES).
- Test and validate the model on other stocks to assess generalizability.

9.2 Open Questions

- How might shape constraints interact with real-time, high-frequency data?
 - Can these models be improved by integrating additional market indicators?
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Conclusion

This study demonstrates that shape-constrained regularization provides a robust alternative to traditional regularization methods for financial volatility forecasting. By enforcing structural constraints, this approach yields more stable and interpretable forecasts, aligning with market trends. Future research should explore the scalability and adaptability of shape-constrained models across various financial metrics and data frequencies.

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